

Solid State Physics Tutorial

Energy Levels, Band Structure, and Effective Mass

1 Hydrogen Atom Energy Levels

1.1 Conceptual Questions

1. **Question 1.1:** What is the physical significance of the Bohr radius a_0 in the hydrogen atom, and how does it relate to the uncertainty principle?
2. **Question 1.2:** Explain why the energy levels of the hydrogen atom are negative, and what does $E = 0$ represent?
3. **Question 1.3:** How does the energy difference between adjacent levels change as we go from $n = 1$ to higher quantum numbers, and what does this tell us about ionization?

1.2 Numerical Exercises

Exercise 1.1: Energy Transition in Hydrogen

Problem: An electron in a hydrogen atom is in the $n = 3$ state. It transitions to the $n = 2$ state by emitting a photon. Calculate:

- (a) The wavelength of the emitted photon
- (b) The energy of the photon in eV
- (c) In which region of the electromagnetic spectrum does this photon lie?

Given Data:

- Rydberg constant: $R_H = 1.097 \times 10^7 \text{ m}^{-1}$
- Planck constant: $h = 6.626 \times 10^{-34} \text{ Js}$
- Speed of light: $c = 3.0 \times 10^8 \text{ m/s}$
- $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Exercise 1.2: Ionization from Excited State

Problem: An electron in a hydrogen atom is in the $n = 2$ excited state. Calculate:

- (a) The binding energy of the electron in this state
- (b) The minimum energy required to ionize the atom from this state
- (c) How many times more energy is needed to ionize from $n = 2$ compared to ionizing from $n = 1$?

Given: The ground state energy of hydrogen is $E_1 = -13.6$ eV

Exercise 1.3: Wave Function Probability

Problem: For a hydrogen atom in the ground state ($n = 1$, $l = 0$), the probability of finding the electron within a distance r from the nucleus is related to the Bohr radius $a_0 = 0.529$.

- (a) What is the most probable radius (position) where the electron is found?
- (b) Calculate the probability of finding the electron within one Bohr radius
- (c) What is the physical significance of the Bohr radius in this context?

Given:

- Bohr radius: $a_0 = 0.529 \times 10^{-10}$ m
- For the $1s$ orbital: $R_{10}(r) = 2a_0^{-3/2}e^{-r/a_0}$
- Radial probability density: $P(r) = |R(r)|^2r^2$

2 Kronig-Penney Model for Band Structure

2.1 Conceptual Questions

1. **Question 2.1:** What is the fundamental difference between the band structure produced by the Kronig-Penney model and a free electron gas, and why does the periodic potential create bands?
2. **Question 2.2:** Explain the concept of Brillouin zones and how they relate to reciprocal space. Why do we only need to consider the first Brillouin zone?
3. **Question 2.3:** What does the band gap represent physically, and under what conditions in the Kronig-Penney model does it become larger?

3 Effective Mass Calculation for Electrons and Holes

3.1 Conceptual Questions

1. **Question 3.1:** What is meant by "effective mass" and why is it different from the free electron mass? Under what conditions can it become negative or infinite?
2. **Question 3.2:** Explain the physical difference between electrons near the bottom of a band and holes near the top of a band, using the concept of effective mass.
3. **Question 3.3:** Why is the concept of effective mass so powerful in semiconductor physics, and what limitations does it have?

3.2 Numerical Exercises

Exercise 3.1: Effective Mass from Band Curvature

Problem:

For a parabolic band approximation, the energy dispersion near the band minimum is:

$$E(k) = E_c + \frac{\hbar^2 k^2}{2m^*} \quad (1)$$

where m^* is the effective mass, E_c is the conduction band minimum, and k is the wave vector measured from the band center.

For a particular semiconductor, measurements give:

- At $k = 0.1 \times 10^{10} \text{ m}^{-1}$: $E = E_c + 0.05 \text{ eV}$
- At $k = 0.2 \times 10^{10} \text{ m}^{-1}$: $E = E_c + 0.20 \text{ eV}$

- (a) Calculate the effective mass
- (b) Express it as a fraction of the free electron mass
- (c) What is the physical meaning of this result?

Exercise 3.2: Holes and Negative Effective Mass

Problem:

Near the top of a valence band in a semiconductor, the energy dispersion is:

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m_h^*} \quad (2)$$

where E_v is the valence band maximum and m_h^* is the hole effective mass. Measurements show:

- At $k = 0$: $E = E_v$
- At $k = 0.1 \times 10^{10} \text{ m}^{-1}$: $E = E_v - 0.08 \text{ eV}$

- (a) Calculate the hole effective mass
- (b) Explain why a missing electron (hole) behaves as a positive charge
- (c) Show that the velocity and force on a hole are opposite to those on an electron

Isotropic vs Anisotropic Dispersion

The effective mass, m^* , describes how an electron or hole responds to an external force inside a crystal. For isotropic materials, m^* is a scalar, but for anisotropic materials, it is a tensor, reflecting the directional dependence of the band curvature. This exercise explores these concepts.

A novel semiconductor material has been synthesized, and its conduction band edge near the Γ -point is described by the following energy-momentum (E - $\mathbf{k}$) dispersion relation:

$$E(\mathbf{k}) = E_c + \frac{\hbar^2}{2} \left(\frac{k_x^2}{m_x^*} + \frac{k_y^2}{m_y^*} + \frac{k_z^2}{m_z^*} \right)$$

where E_c is the energy at the conduction band minimum, $\hbar$ is the reduced Planck constant, and m_x^* , m_y^* , and m_z^* are the effective masses along the crystallographic axes. The material's properties are determined to be $m_x^* = 0.2m_0$, $m_y^* = 0.5m_0$, and $m_z^* = 0.3m_0$, where m_0 is the free electron mass.

Questions

1. **Effective Mass Tensor:** Write down the 3×3 effective mass tensor, $\mathbf{m}^*$, for this material.
2. **Acceleration in Response to an Electric Field:**
 - (a) An electric field, $\mathbf{E} = E_0\hat{\mathbf{x}}$, is applied along the x -direction. Calculate the electron's acceleration vector, $\mathbf{a}$. Is the acceleration parallel to the applied force?
 - (b) Now, an electric field, $\mathbf{E} = E_0(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, is applied. Calculate the electron's acceleration vector, $\mathbf{a}$. Is the acceleration parallel to the applied force?
3. **Comparison with an Isotropic Material:**

A second, isotropic semiconductor has a single, spherical conduction band valley at Γ . This material has an effective mass of $m^* = 0.2m_0$. An electric field, $\mathbf{E} = E_0(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, is applied.

 - (a) Calculate the electron's acceleration vector, $\mathbf{a}$.
 - (b) Compare this result with your finding in Question 2b. Explain the difference in the acceleration vectors based on the material's isotropy versus anisotropy.

Key Formulas

$$\left(\frac{1}{\mathbf{m}^*} \right)_{ij} = \frac{1}{\hbar^2} \frac{\partial^2 E(\mathbf{k})}{\partial k_i \partial k_j}$$

$$\mathbf{a} = \left(\frac{1}{\mathbf{m}^*} \right) \cdot \mathbf{F}, \quad \mathbf{F} = -e\mathbf{E}$$